DeepLense Unsupervised Super Resolution and Analysis of Real Lensing Images

Google Summer of Code 2026 Proposal for Machine Learning for Science (ML4Sci)

Applicant Information

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1. Executive Synopsis

Figuring out the exact nature of dark matter is one of the biggest challenges in modern physics. Strong gravitational lensing gives us a rare way to actually "see" how dark matter is distributed on subgalactic scales without needing to rely on light or electromagnetic emissions. The problem is that ground-based telescopes suffer from atmospheric blurring, and getting time on highly precise space-based observatories is incredibly expensive.

For my 350-hour GSoC project, I want to solve this hardware bottleneck. I will build and train a fully unsupervised score-based generative super-resolution model. By utilizing Noise Conditional Score Networks (NCSN++), my architecture will learn the physical rules of high-resolution lenses from simulations, and then upscale blurry, real-world observations without needing paired ground truths.

2. Astrophysical Context & Community Benefits

We know the standard ΛCDM model works incredibly well on a massive cosmic scale, but it starts showing cracks on subgalactic levels (like the missing satellites problem). Newer theories suggest dark matter might behave like a superfluid, creating quantum vortices. To tell the difference between standard spherical dark matter subhalos and cylindrical vortices, we need to observe microscopic gravitational deflections.

Because getting empirical, high-resolution data is so hard, the community relies heavily on synthetic datasets (like PyAutoLens). But as we know, models trained purely on simulations often

break down when exposed to real-world domain shifts. Inspired by recent advancements in Image-to-Image translation in astronomy, my unsupervised pipeline will bridge this "simulated-to-real" gap. It will give researchers a tool to confidently upscale existing, abundant low-resolution data, and help prepare the DeepLense pipeline for the massive wave of data coming from the Vera C. Rubin Observatory (LSST) and Euclid.

3. DeepLense Heritage & Technical Approach

3.1 Building on Previous Architectures

I've spent time reading through the mentors' previous research, and my project is designed to combine the best parts of their work:

- **Classification (Alexander et al.):** Proved that CNNs can decode dark matter substructures with high accuracy.
- **DiffLense (Pranath Reddy):** Showed that conditional diffusion models easily beat standard SRCNNs at preserving physical details. However, it relies on perfectly paired data, which we don't have for real telescope images.
- **LensSR (Anirudh Shankar):** Showed the massive value of physics-informed, unsupervised super-resolution.

My Approach: During the ML4Sci evaluation phase, I built a supervised SRCNN with a custom structural loss function that achieved a 41.32 dB PSNR. Standard MSE-based models tend to blur out microscopic anomalies, so I engineered an α-blended objective:

$$\mathcal{L} = \alpha \cdot \text{MSE}(I_{SR}, I_{HR}) + (1 - \alpha) \cdot (1 - \text{SSIM}(I_{SR}, I_{HR})) \tag{1}$$

While this deterministic baseline is strong, to actually hallucinate missing details from unpaired real-world datasets, we need to move to Score-Based Generative Models (SDEs), heavily drawing from Yang Song's foundational research.

3.2 Unsupervised Inference via Predictor-Corrector

I plan to train the NCSN++ model unconditionally to approximate the true score function using reverse-time Stochastic Differential Equations. When we feed it a low-resolution observation (degraded by atmospheric blur and sensor noise), it will sample from the posterior to generate the high-resolution equivalent. I will implement the Predictor-Corrector algorithm to handle this:

1. **Predictor Step:** Using the Euler-Maruyama method to simulate the reverse-time SDE.
2. **Corrector Step:** Using Langevin MCMC dynamics to add calibrated noise, ensuring the image stays physically accurate.
3. **Data Consistency:** Projecting the generated image so that if we applied our atmospheric blur operators to it again, it would perfectly match the original observation.

4. Project Deliverables and Implementation Timeline (350-Hour Scope)

This will be my first GSoC, and I am incredibly excited to dive in. I have cleared my summer schedule and am fully prepared to commit 30–35 hours per week to hit these milestones. I have intentionally marked every phase of this architecture as [**Required**]. I do not view any component of this pipeline as optional; I am ready to tackle the full scope of the project and will treat any additional challenges or directional shifts from the mentors as core mandates.

Phase 1: Generative Prior (Weeks 1–3)

- [**Required**] **Investigation:** Deep dive into Langevin dynamics and SDE-based diffusion generative models.
- [**Required**] **Coding:** Modify data loaders for an unsupervised training setup, handling real HST empirical tensors.
- [**Required**] **Coding:** Implement the baseline unconditional NCSN++.

Phase 2: Domain Adaptation & Unsupervised Inference (Weeks 4–7)

- [**Required**] **Coding:** Write degradation operators to simulate the atmospheric blurring and sensor noise of real ground-based telescopes.
- [**Required**] **Coding:** Build the Predictor-Corrector loop (Euler-Maruyama + Langevin MCMC).
- [**Required**] **Coding:** Integrate mathematical data consistency steps to ensure outputs remain physically viable.

Phase 3: Substructure Analysis Module & Metrics (Weeks 8–10)

- [**Required**] **Coding:** Take the multi-class classification pipeline I built for the evaluation task (>0.98 AUC) and port it as a downstream tool to verify the shapes of our generated images.
- [**Required**] **Investigation:** Set up Fréchet Inception Distance (FID) to benchmark the generative outputs against my supervised baselines.

Phase 4: Wrap-up & Handover (Weeks 11–12)

- [**Required**] **Documentation:** Clean up the codebase (PEP8 standards) and refactor for GPU memory efficiency.
- [**Required**] **Deliverables:** Write clear Jupyter Notebook tutorials so future researchers can actually use my code, merge final PRs, and submit the GSoC report.

5. Biographical Information and Technical Qualifications

I am currently a Third Year Undergraduate and an AI/ML Engineering Intern at ITAC (OSU) Oklahoma remotely, executing a project funded directly by the **U.S. Department of Energy**. I am building a highly secure, air-gapped Zero-Trust RAG pipeline. Working within federal infrastructure is teaching me how to write robust, production - ready code - skills I will bring directly to handling DeepLense's complex datasets.

Before this, I was a Generative AI Intern at Canspirit AI, where I optimized spatial and visual AI workflows. I also love building things from scratch. My independent projects include writing a low-latency, lock-free Pub-Sub messaging broker in C++ to eliminate heap fragmentation. I am comfortable diving into deep technical architectures.

Finally, I know how to manage projects and communicate. As the **Chairperson for Google Developer Groups (GDG)**, at VIT Pune; I lead a community of over 125 developers. I recently organized our campus's largest hackathon and managed the launch of Asia's first Liquid Galaxy Lab. Since Liquid Galaxy is also a GSoC umbrella organization, I'm very familiar with the open-source community ethos, have a great repo with many GDE's, documentation standards, and also the work ethic required to succeed here. I am confident I have the right mix of backend engineering, ML research experience, and project management skills to deliver a great result for ML4Sci this summer.